Tail Analysis Indicates Lost Seals

immediate

Abstract **Complex polygenic traits are typically modeled as the additive sum of many variants of limited effect alongside rare risk alleles of substantial penetrence. This distinction likely reflects a balance between long-range selection pressure and the limited ability of trait-altering mutant alleles to persist over multiple generations. Identification of individuals with extreme trait values as probable carriers of rare high-impact variants has led sequencing strategies to focus on individuals in the tails of the trait distribution. Tail trait values can also result from extreme polygenic burden created by recombination and reassortment. Therefore, to enhance the efficency of sequencing studies, it is imperative to be able to discriminate between these different etiologies.**

One indicatation that rare variants may play a significant role in the tails of the trait distribution can be reflected in the reliability of polygenic risk scores (PRS) in the tails. If highly penetrant rare variants not included in GWAS determine the tails of the trait distribution such individuals will not be assigned appropriate risk scores. In this study, we develop a *polygenic regression-to-mean* (PRM) statistical test that compares the PRS in the tails of the trait distribution to expectation PRS under a polygenic model. Applying this test to hundreds of traits in the UK Biobank reveals that the majority of quantitive traits exhibit significant PRM in one or both tails.

Next we examine whether rare variants in exome studies are enriched in trait tails with significant tail regression and demonstrate that such tails are more likely to be under negative selection. Further examining these trait tails we find an association between PRM and other measures of fitness that may be influenced by rare variants including the lifetime number of illnesses and still births. We then compare these results to a simulation and find that they match.

Finally, we assess whether the predictions of a complex trait architecture derived from a conditional sibling model align with the observed diversity in the architecture of trait extremes. The concordance between these measures underscores the need to consider the role of rare variants in the analysis of trait tails. The PRS and sibling analyses presented here offer valuable tools for the targeted selection of traits and samples likely to harbor de novo or rare deleterious alleles and represent an approach that has the potential to significantly enhance the effectiveness and design of sequencing studies.

Introduction

Genome-wide association studies (GWAS) are instrumental in understanding the genetic architecture of complex traits. These studies typically identify numerous common genetic variants, each of small individual effect (1) that can be used in summation to make predictions. The idea that these common variants adequately represent the genetic architecture is supported by evolutionary theory, which suggests that many human traits (e.g., height, birth weight) are under stabilizing selection, which reduces the fitness impact of large-effect variants (2).

A recent hope for advancing personalized medicine is in the potential of polygenic risk scores to identify high-risk individuals who may benefit from early interventions, such as adoption of specific lifestyle regimes or increased screening (3–6). However, this strategy will be underpowered when applied to traits or diseases with reduced PRS in the tail of the distribution due to reduced common polygenicity. To overcome this problem, the genetic aetiology underlying the tails of phenotype distributions must be investigated and better characterized (7). Polygenic prediction models use these common small-effect variants to generate polygenic risk scores (PRS) whose performance making out-of-sample predictions is usually evaluated with aggregate estimates like *variance explained* (R^2). However, new evidence has suggested that this level of analysis may not be adequate because the preformance of prediction models is not uniform across the trait distribution. This

non-uniformity results because the contribution of small effect common variants relative to highly penetrant rare variants and the environment is not equal across the trait distribution (7, 8).

This negative correlation between effect size and minor allele frequency (MAF) reflects the interplay between long term selection against extreme trait values and the ability of mutant alleles with significant effects to escape this trend through chance or short term environmental changes. Recent or ongoing selection can leave distinct genetic signatures, such as an enrichment of rare variants in individuals in the tails of trait distributions during directional or stabilizing selection. The expectation that individuals with extreme trait values are more likely to carry rare large-effect variants has led to strategies that priorize sequencing of individuals in the tails of the trait distribution (9, 10). However, tail trait values can also result from extreme polygenic burden created by recombination and reassortment, and a more efficient approach would involve considering more lines of individuals to allow the selection of individuals with strongest evidence of rare variant architecture for sequencing.

Here we attempt to distinguish whether the tails of the distribution of different traits are primarily driven by common polygenicity or by a more "monogenic-type" architecture **NEED: monogenic literally means single-gene disorder**. We accomplish this by developing a statistical test that measures the fit of the polygenic model across the trait-PRS distribution and identifies non-polygencity in each tail and applying it to over a **NEED: thousand** UK Biobank traits (see ***Figure 1***) **NEED: sentence is too long and "trait-PRS distribution" is inaccurate description**. Next we consider whether evidence of negative selection exists in a subset of uncorrelated traits **NEED: we use all 75 traits, these are not uncorrelated** and show that the trait tails under negative selection are more likely to be associated with **NEED: PRS regression to the mean which may imply existence of rare variants**. We further triangulate evidence for selective constraint leading to reduced contribution of common polygenicity in the tails, by testing for the associations between known measures of fitness and PRS regression to the mean in the tails. (11, 12).

Finally, we assess whether the predictions of a complex trait architecture derived from a conditional sibling model align with the observed diversity in the architecture of trait extremes inferred from population level analyses.

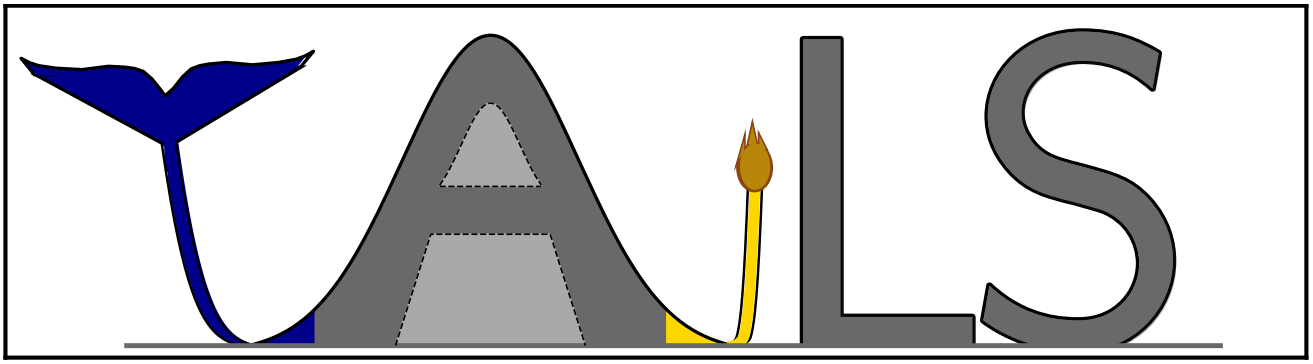


Fig. 1: **Placeholder for Schematic**

Results

Population-level data analyses for rare variants of large effect affecting the tails of trait distributions.

We use PRS to assess the contribution of rare large-effect variants on extreme trait values. Rare variants are typically not included in PRS, leading to reduced accuracy of the PRS for individuals with rare large-effect variants. This inaccuracy will be most apparent in the tails of the trait distribution where individuals with large-effect variants are most likely to reside. Specifically, PRS values will tend to converge toward the mean PRS in the tails – *PRS-regression to the mean*. To quantify this we develop and apply a statistical test that estimates effect size, direction, and signficance of movement away from the expected PRS for individuals in the lower and upper 1% of the trait distribution. Specifically, a regression of PRS on standardised trait values is fit. The regression effect size in the tail is defined as the mean of the residuals in the lower tail and minus the mean of the residuals in the upper tail. Using this definition, positive values in both the upper and lower tails correspond to regression to the mean, indicative of rare variants of large effect in individuals in the trait tails. **NEED: Negative values correspond to regression away from the mean.....**

We first apply our test to UKB samples of European ancestry. Available samples were split equally, one half of the data was used for GWAS to estimate PRS via PRSice and our PRS-test was applied to the remaining samples. Throughout, PRS were estimated with variants with MAF $> 1\%$, see Methods for further details. Unimodal, non-skewed, continuous traits with heritabilty > 0.05, PRS $R^2 > 0.01$ and sample size >50,000 were selected for analysis. **NEED: Also removed 3 traits with single variants with $r^2 > 0.5$, is this necessray? Could just group and discuss with other two traits which fail PRS-test QC**. After removal of strongly correlated traits (maximum pairwise correlation R^2 <0.95) 77 traits remained for analysis. Traits were standardised, regressing out effects of age, sex, recruitment centre, genotyping batch, and principal components and transformed to be normally distributed, see Methods for further details of trait QC and selection. Of these traits, two failed our test QC indicating significant departure from predicted values from the regression in the middle 90% or the trait distribution (see Methods) leaving 75 traits (150 tails) for analysis. Extended Data Table 1 lists all 77 traits and their available sanple sizes.

Figure 2a illustrates the three broad types of tail regression by plotting trait quantile, in 1% increments, versus mean PRS in the quantile: no tails regression exhibited by triglycerides (chosen as the trait with highest h^2 of any trait without tail regression to the mean), only lower tail regression to the mean (haemoglobin concentration), only upper tail regresion to the mean (red blood cell distribution width) and dual tail regression to the mean (sitting height). Across the 75 traits analysed, the mean PRS in 121 of 150 upper and lower 1% tails of trait distributions showed significant deviation (FDR<0.05) away from the expected PRS. Of these 121 significant tail associations, 118 were regression to the mean. ***Figure 2b*** shows the PRS-test tail effect sizes for all traits with significant regression to the mean in at least one tail. The majority of traits (47 of 75) were classified as "Dual Tail" (FDR<0.05 regression to the mean in both tails), 15 were classified as "Lower Tail" and 9 were classified as "Upper Tail". **NEED: The three trait tail with significant regression away fromn the mean... The two traits failling test QC.....**

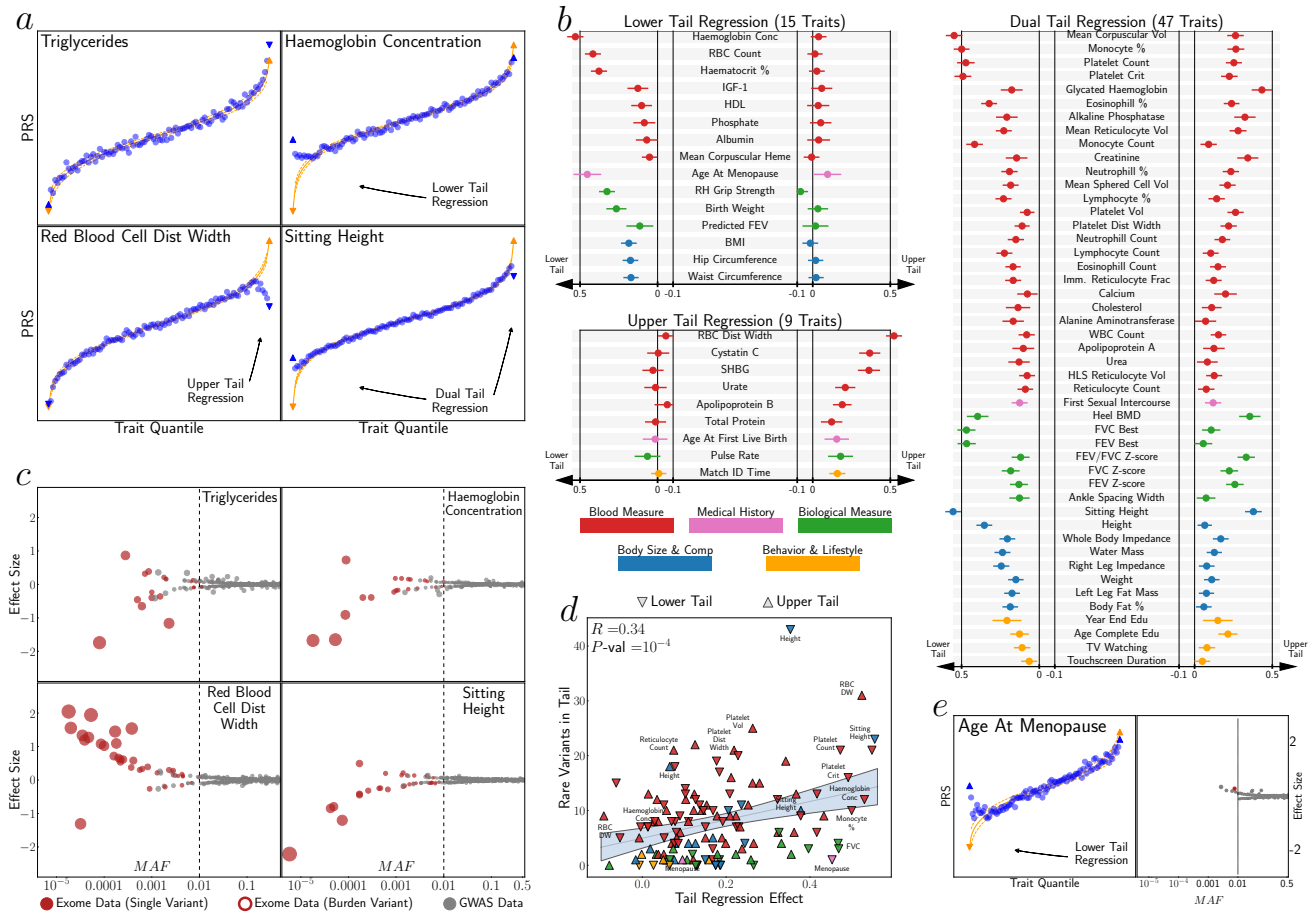
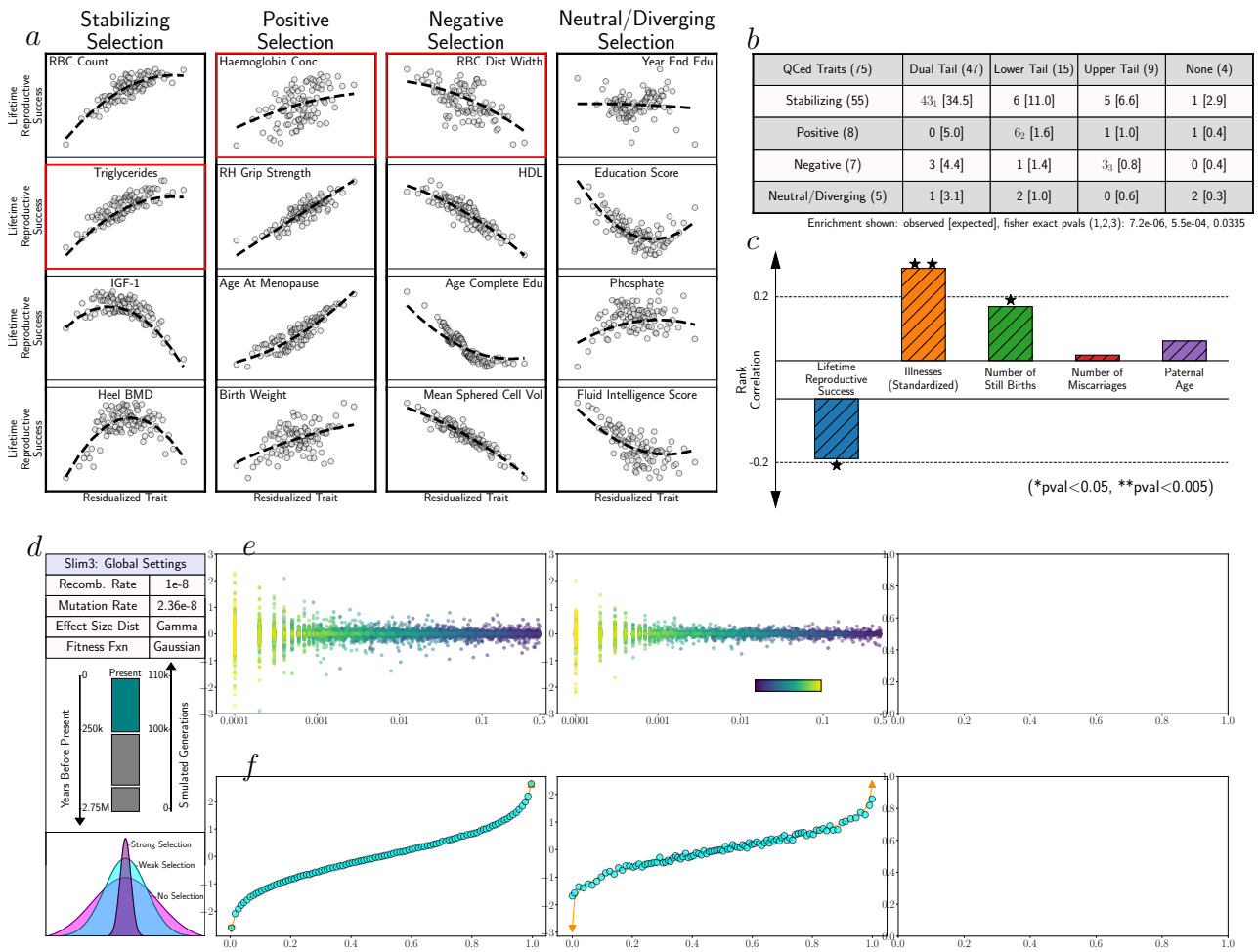


Fig. 2: **Population-level data analyses for rare variants of large effect affecting the tails of trait distributions.** **(a)** PRS quantile (in 1% increments) plotted agaisnt mean trait value in quantile for four traits with different types of PRS tail regression (clockwise: No Tail Regression, Upper Tail, Lower Tail, Dual Tail). **(b)** PRS-test tail effect sizes and **NEED: 95% confidence intervals** in the upper and lower 1% tails for all traits (of the 75 passing QC) which exhibit significant regression to the mean in one or both tails at FDR of 0.05. **(c)** Trumpet plots (minor allele frequency v effect size) of genome-wide significant variants for traits shown in (a). **(d)** PRS-test tail effect sizes plotted against number of genome-wide significant variants with MAF<1% for X traits with overlapping exome sequence results. For upper tails positive variant effects are counted and for lower tails negative variant effects are counted. Correlation and significance calculated using Spearman rank correlation. **(e)** Age of menopause: PRS quantile v trait mean and trumpet plot.

The Role of Rare Variants in PRM

The exclusion of rare variants from PRS (traditionally a cutoff of minor allele frequency (MAF)>1%) has been shown to reduce model performance (13). To investigate whether PRM is a consequence of rare variants of large effect that drive individuals into the tails of the trait distribution we created trumpet plots (***Figure 2***c) to visualize the impact and direction of of rare variants from GWAS (14) and exome studies (15). Focusing on our single tail index traits, we notice that there are a greater number of rare variants in the lower tail of *Haemoglobin Concentration* and in the upper tail for *Red Blood Cell Distribution Width*. Additionally, for *Sitting Height* there are more rare variants in the lower tail (where the PRM Effect is greatest) and in *Triglycerides* there are also more rare variants in the lower tail (where PRM Effect is greatest, albiet non-signficant). This result is consistent with the hypothesis that rare variants of large effect that not accounted for in the common-variant PRS responsible for the tail specific PRM.

In ***Figure 2***d we observe a small but signficiant relationship between the magnitude of regression to the mean and the number of rare variants in the tail. However, for some traits the tail with more PRM shows almost no rare variants (see ***Figure 2***e) suggesting that strong selection may result in private variants that cannot be identified using population based sequencing.



QCed Traits (75)	Dual Tail (47)	Lower Tail (15)	Upper Tail (9)	None (4)
Stabilizing (55)	43_1 [34.5]	6 [11.0]	5 [6.6]	1 [2.9]
Positive (8)	0 [5.0]	6_2 [1.6]	1 [1.0]	1 [0.4]
Negative (7)	3 [4.4]	1 [1.4]	3_3 [0.8]	0 [0.4]
Neutral/Diverging (5)	1 [3.1]	2 [1.0]	0 [0.6]	2 [0.3]

Enrichment shown: observed [expected], fisher exact pvals (1,2,3): 7.2e-06, 5.5e-04, 0.0335

Fig. 3: Selection analyses in real and simulated data. (a) Residualised trait value plotted against relative lifetime reproductive success for 16 **NEED: cherry picked traits**, quadratic best fit to the data is shown by dashed lines. **(b)** Number of traits with inferred type of PRS tail regression and inferred type of selection for 75 UKB traits passing QC. Tail regression is inferred if tail effect is significant at an FDR of 0.05. **(c)** Spearman rank correlation between standardised mean proxy measures of fitness of individuals in the upper and lower 1% tails and PRS-test effect size for the corresponding tail across 75 traits in UKB passing QC. **(d)** **NEED: not sure this is needed**. **(e)** Trumpet plots (minor allele frequency v effect size) for simulated data without and with selection. All variants are included in the plot. **(e)** PRS quantile (in 1% increments) plotted agaisnt mean trait value in quantile in simulated data without and with selection. PRS calculated from all variants with MAF>1%.

The Impact of Selection on PRM

Using adjusted lifetime reproductive success (2, 16) as a proxy for selection we classify (see Methods - LRS) each trait with respect to whether it is under *Stabilizing, Positive, Negative, or Neutral/Disruptive* selection (these results are in agreement with a more comprehensive sex-specific analysis (See ***Figure 9***). The majority of traits analyzed here (55 of 75) are classified as under stabilizing selection which is characterized by lower fecundity for individuals with traits values in either tail of the trait distribution. Stabilizing selection may be the result of general fitness favoring the trait mean or the result of sex specific selection that averages to produce this relationship (2). Eight traits were classified as being under positive selection and seven traits were classified as being under negative selection. We also classified four traits as under neutral selection (no signficant relationship with LRS) and one trait as divergent/disruptive selection where fecundity is higher in both tails that at the mean of the distribution (***Figure 3***a). We find a significant relationship between selection classification and PRM classifition (***Figure 3***b), whereby traits under stabilizing selection are enriched in "Dual Tail Regression" ($p < 2.5 \times 10^{-5}$), traits under positive selection are enriched in "Lower Tail Regression ($p < 2.4 \times 10^{-4}$), and traits under negative selection are enriched in "Upper Tail Regression ($p < 0.024$). Focusing on each tail separately we find an even stronger relationship insofar as the in the 118 trait tails which exhibit PRM, the tail is selected against 110 times (93%) while in the 32 tails which do not exhibit PRM the tail is only selected against 16 times (50%) ($pv < 0.001$).

Tail Fitness

In addition to *lifetime reproductive success*, we analyzed four measures of fitness that are likely to be associated with reduced fitness and rare variant or *de novo* etiology. Here, we note that under the hypothesis that rare variants under negative selection are responsible for PRM, we predict that trait tails with large effect PRM will have lower average *lifetime reproductive success* and higher average numbers of *Still Births* and *Miscarriages*. Additionally, traits tails with large effect PRM are predicted to have more standarized illnesses because both result from large effect variants, although it is possible that downstream effects of illnesses results in extreme trait values. Finally, if paternal age is associated with more *de novo* mutations (17) it should also be associated with PRM.

In ***Figure 3***c the rank correlation for the average tail fitness measure and the PRM effect size across the trait tails is shown. We observe that the direction of rank association is in line with the rare-variant hypothesis in every case, although the association between PRM and the number of miscarriages and paternal age are not significantly associated.

Simulating Selection (Anil's Result)

In ***Figure 3***d we confirm this result using a simulation (see *Methods*) where we simulate many generations of stablilzing selection, after which the ratio of contribution to heritability by rare variants goes up and PRM is observed.

Replication in Siblings

Discordance between siblings in the tails of the trait distribution is indictive non-polygenic inheritance (18). Furthermore, excess trait concordance between siblings can be indicative of shared Mendelian effects (19). ***Figure 4***a illustartes the expected impact of polygenicity, de novo, and Mendelian rare variants within sibling pairs. Expected sibling pair concordance under polygenicity can be used to test for (1) excess discordance between siblings in tails of trait distribution, indicative of de novo mutations of large in one sibling and (2) excess concordance between siblings in tails of trait distribution, indicative of shared Mendelian effects (19). These tests are derived from the conditional sibling distribution under polygenicity, this distribution also allows heritability to be estimated (19).

Applying this model to the sibling data in the UKB to estimate h^2, we find a correlation between snp h2 and sibling h2 ***Figure 4***b. In our index traits we use this model to infer the presence of rare variant architecture in the tails and find that lower tail discordance in *Haemoglobin Concentration*, upper tail discordance in *Red Blood Cell Distribution Width* and dual ail discordance in *Sitting Height*, all of which match the result from our PRM test. Overall we find a significant relationship between our sibling test pvalue and the pvalue from our PRM-test ***Figure 4***d.

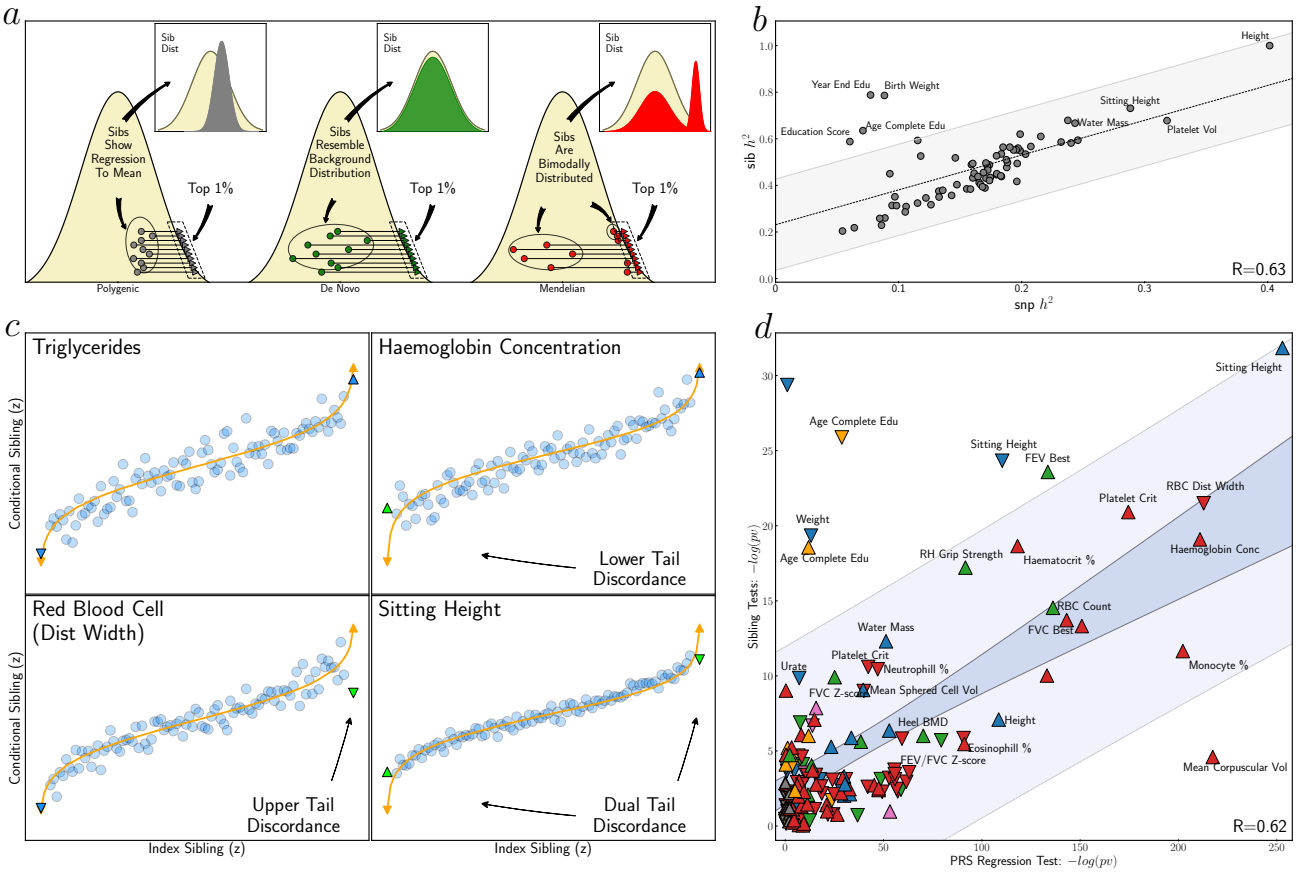


Fig. 4: **Sibling analyses for rare variants of large effect affecting the tails of trait distribution.** **(a)** Schematic of relationship of sibling trait values under polygenicity, de novo variants of large effect in one sibling and rare inherited variant of large effect, Mendelian. **(b)** SNP heritability estimtaed from *LDscore applied to UKB* plotted against sibling heritability estimated from UKB sibling pairs. **(c)** Index sibling trait quantile (in 1% increments) plotted agaisnt mean trait value of conditional sibling in quantile for four traits with different types of PRS tail regression. **(d)** PRS-test P-values plotted against sibling meta test P-values on $-\log_{10}$ scale for upper and lower tails of X traits. Correlation **NEED: and significance** calculated using Spearman rank correlation.

Replication

We performed three *flavours* of replication.

1. **Repeated Subsets of The Target Data:** Assess the reliability of our PRM-test in smaller samples
2. **Non-European Ancestry Inviduals in UKB:** PRM replication across ancestry (within biobank)
3. **European Ancestry Individuals in All Of Us:** PRM replication across biobank (within ancestry)

Bootstrap Analysis using Target Subsets

To assess the performance of our PRM-test using smaller samples we re-sampled small subsets of our target populations, and performed our PRM-test on these subsets. In 5d we notice that even the correlation of effect sizes was approximately 0.70 even when the subset size was only 5k, however concordance in discovery was poor at these levels.

Out of Sample Replication

The correlation between PRM effect-sizes was greater in the UKB Non-European Ancestry data despite smaller sample sizes, although the number of true positives identified was lower (underpowered). This suggests that even if PRS isn't portable across ancestry, that regression to the mean is portable and in fact may be even more than portable.

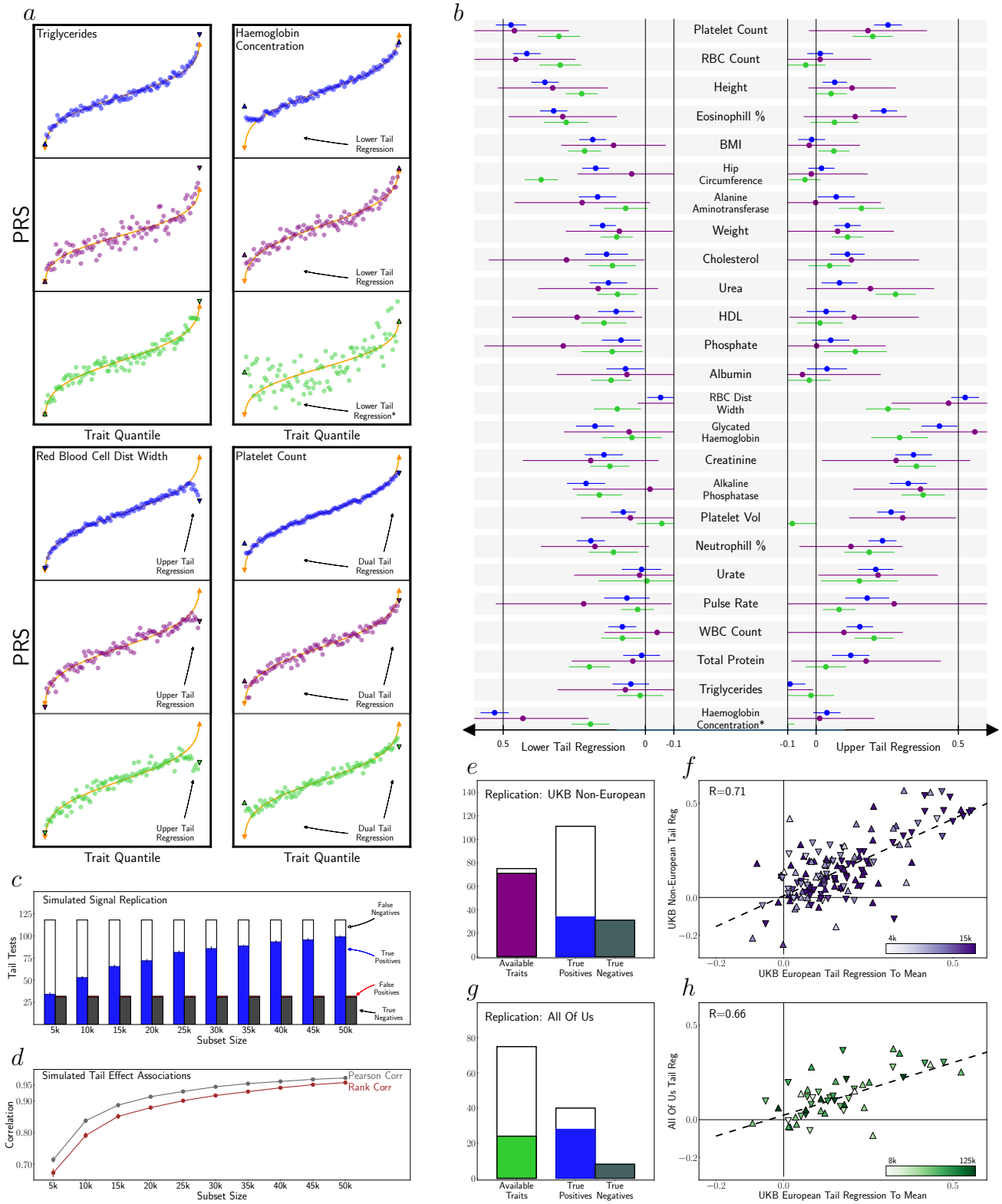


Fig. 5: Replication results in UKB non-European samples, All of Us European ancestry samples and bootstrap analyses of UKB European samples. (a) PRS quantile (in 1% increments) plotted agaisnt mean trait value in quantile in UKB European ancestry samples, UKB non-European ancestry samples and All Of Us European ancestry samples for four traits with different types of PRS tail regression. **(b)** PRS-test tail effect estimates in UKB European ancestry samples, UKB non-European ancestry samples and All of Us Euroepan ancestry samples for 25 traits measured and passing PRS-test quality control in the three cohorts. **(c)** Simulated Signal Replication: Increasing larger subsets were sampled and the replicability relative to our UKB result was calculated **(d)** Same thing but correlation. **(e)**. **(f)** PRS-test effect sizes estimated in: UKB European ancestry samples v UKB non-European ancestry samples for X traits passing PRS-test QC in both cohorts. Triangles shaded relative to the non-European sample size for the trait. Pearson correlation shown**(g)**. **(h)** PRS-test effect sizes estimated in: UKB European ancestry samples v All Of Us European ancestry samples for 24 overlapping traits passing PRS-test QC in both cohorts. Triangles shaded relative to the All of Us sample size for the trait.

Discussion

In this study, we propose the use of PRS Regression-to-the-Mean (RM) Test as a way to identify traits that are enriched for rare or de novo – more "monogenic-type" effects (MTE) in extreme. We applied this method to a list of quantitative traits available in UKB. In addition to genotype data, we also incorporated the rich source of UKB information including the sibling data (Sibling RM Test), and paternal age (Paternal Age Test) and number of children (Fecundity Test) into our analyses. We found directional consistency in most of the tests performed, providing converging evidence for the feasibility of using PRS as a strategy to identify traits with potential MTE. In our analyses, the PRS RM test can be used to identify both the "polygenic" and "MTE" event, we hypothesize that for a highly polygenic trait, samples with extreme trait values (ETV) are expected to have extreme PRS correspondingly. Deviation from the expectation may suggest samples at the extreme tails are enriched for MTE that are under selection. Our results show significant correlation in the ranking traits from least to most evidence of common polygenicity in the tails of their distributions between the PRS RM test and Fecundity test. Furthermore, the results of the PRS RM test are supported by the Sibling RM test of which we use the sibling phenotype data to investigate the common polygenicity in different traits. Sib-pairs are expected to have similar trait value under the influence of common polygenicity. On the other hand, if the ETV in sib A is contributed by de novo mutations, we expect sib B to be discordant phenotypically. An alternative explanation for discordant phenotype in sib-pairs could be due to the non-shared environment factors with large effect.

The striking concordance between these measures underscores the need to consider the role of rare variants in the analysis of trait tails. The PRS and sibling analyses presented here offer valuable tools for the targeted selection of traits and samples likely to harbor de novo or rare deleterious alleles and represent an approach that has the potential to significantly enhance the effectiveness and design of sequencing studies.

One major limitation of these tests is that they are highly sensitive to measurement errors. Measurement errors would result in the same PRS/ sibling departure at the extreme ends, because the PRS is no longer reflective by one's phenotypic measurement. In addition, both the PRS RM and sibling RM Test cannot distinguish between large effect mediated by environmental factors and MTE. For example, if a person with extremely low levels of LDL but normal PRS can be caused by malnutrition. Therefore, collection of the associated environmental factors and the corresponding adjustment becomes vital to alleviate this problem when applying these tests. Moreover, these tests are only applicable to quantitative traits, more work is needed to develop for testing binary outcome. In this study, we identified traits with evidence to be influenced by MTE at the trait tail(s), however, we did not identify or confirm the corresponding de novo mutation. Despite the lack of parent-offspring trios which is essential for the identification of de novo mutations, it is technically challenging to define a list of candidate genes and variants for every trait we investigated. Further investigation is warranted to validate this finding. In conclusion, we propose a novel way of using polygenic risk score to look for evidence of de novo/ rare mutations. This can be used as a method to identify traits that are more likely influenced by de novo mutations, and select samples who are more likely carry de novo mutations, potentially maximizing the power of a sequencing study. This method can be applied to any traits with continuous measure such as age of onset, disease severity and rate of disease progression.

Methods

Data Processing

UKB genetic data

Issues:

- More specifics on the numbers.
- Explain ancestry clustering more specifically (which ancestries?
- What PC values chosen?). Nobody could repeat this accurately as written.
- Some loose language in places. Several places where citations needed e.g. "recommended by the UK Biobank team." i
- Need some better context - e.g. In the primary analyses in unrelated individuals (Figures 2, 3, 5)...
- In the analyses performed on siblings (Figure 4)...
- Not sure if Mixed ancestry results needed anymore?

The UK Biobank is a prospective cohort study that included approximately 500,000 participants with measurements of 4,000 phenotypes, recruited across the United Kingdom from 2006 to 2010 (20). The genetic dataset consists of 488,377 samples genotyped at 805,426 SNPs. To determine population ancestries, 4-means clustering analysis was conducted on the first two Principal Components (PCs) of the genotype data. Each cluster's ancestry was determined based on the majority of individuals' country of birth (field ID: 20115) within that cluster.

Subsequent to clustering, standard quality control (QC) procedures were applied independently to each ancestry cluster. SNPs with a minor allele frequency < 0.01, genotype missingness > 0.02, or Hardy-Weinberg equilibrium test P-value $< 10^{-8}$ were excluded. Samples exhibiting high levels of missingness or heterozygosity, inconsistencies in genetic-inferred and self-reported sex, or displaying aneuploidy of the sex chromosomes were removed in accordance with recommendations from the UKB data processing team (21). A greedy algorithm was employed to remove related individuals that maximizes samples retention while removing all third-degree relatives (kinship coefficient > 0.044) (22). After this QC there 432k unrelated individuals remained. Primary analyses were performed on over 410k unrelated European samples and replication analyses was performed on the remaining 22k samples classified as non-European ancestry.

For the sibling based analyses sibling pairs were selected by first identifying first degree relatives defined as those with kinship coefficient within the range 0.177 and 0.354, this range either side of the expected value of 0.25 was used to account for the expected variation in within-pair similarity (23). The proportion of SNPs with zero identity-by-state (IBS0) was used to distinguish sibling pairs from parent-offspring pairs since parent–offspring pairs have IBS0≈0. It has been shown in the UKB there is good separation between pairs of first degree relatives at IBS0=0.00112(23) thus sibling pairs were selected as those that exceeded this threshold, this gave a total of 17,869 sibling pairs (17,289 of European ancestry, that were used in our primary analysis).

Quantiative Trait QC

Issues:

1. Seems a bit unusual to describe the QC, then the residualisation/normalisation, and then some further QC steps on the untransformed data. Is this the best order?
2. I think the reduction from 151 to 92 traits due to rg<0.99 cut-off is likely to raise alarm bells - how many more would've been lost if you'd used 0.98, 0.95, 0.9 instead?
3. There are a lot of P-values reported in the paper dependent on a test of many traits.. but if those are highly correlated then the overall P-value reported will be inflated. We've talked about this before I know but isn't this still a problem!?

Solutions:

1. I don't know. But basically Clives PRM-QC finds non-linearities but we know our normalization can hide them, so it does make sense to be conservative here.
2. Filtering on Clives QC first and doing genetically correlations last (as we should), it hardly makes a difference, I changed it to 0.95 and gave some numbers that should calm alarm bells.
3. They are not highly correlated - but I think we have done enough in the methods to make people calm.
4. This does need to be talked about in the discussion - the whole point of the paper is TAILS, and the tails are nothing like the main part of the distribution, so highly correlated body doesn't mean much about the tails.

Quantiative traits were selected in UKB as those with at least 10 distinct values, heritability $h^2 > 1\%$ (measured by LD score (24) applied to publicly available UKB GWAS summary statistics (14)) and >10,000 observation in unrelated samples, this gave 1,158 traits. Traits values more than six standard deviations from the mean were removed. Traits were then standardised as follows: (1) linear regression models were applied to each trait, with covariates for age, sex, recruitment centre, genotyping batches, and the first 40 genetic principal components, (2) standardised values were obtained by inverse normal transforming the rank of the residuals.

Samples were then split randomly into 50% base and 50% target data sets and GWAS and PRScice were applied and our PRM-regression test was ran on all 1,158 continuous traits (see Supplementary Material). Strong association between target sample size and nominal PRM signficance ($R = 0.5$, p-value$< 10^{-70}$), suggests that many of our tests were underpowered. For this reason we focused all primary analyses on the 136 traits with 50k target samples and PRS $R^2 > 1\%$. (Furthermore, our test assumes that transformation to continuous Gaussian distribution is appropriate and that the underlying genetic architecture is primarily polygenic, therefore we impose the following additional requirements:

- **Continuity:** top two modal untransformed trait values include less than half of samples. (2 traits removed)
- **Unimodal:** Dip-Test ((25)) P-value > 0.05 applied to untransformed trait values. (1 trait removed)
- **Non-skewed:** skew of untransformed trait values <2. (7 traits removed)
- **Polygenic:** Maxmimum single variant $r^2 < 0.5\%$. (3 traits removed)

This futher thinned our traits to 123. To ensure a non-redundant survey of traits, pairwise genetic correlation were calculated using LDscore(24) and 61 traits did not have a pair with $r_g > 95\%$, average top match = 0.60,standard-deviation = 0.2) and kept in the analysis. Additionally, 10 traits has a single almost identical match (eg: "Standing Height" (ID:50) and "Height Measured Prior to Imaging" (ID:12144)), in these cases the trait with greater sample size was priotized. The other 52 traits were part of larger clusters of genetically similar traits, the largest of which related to measurements related to of body-mass (eg: body fat percentage, leg mass, arm mass, etc) and heel bone mineral density. Using the same data maximization algorithm applied to extra unrelated individuals (22), we were able to select 11 of these traits which resulted in a set of 77 genetically non-redudant traits.

Non-European Ancestry Individuals in the UKB

Non-European Ancestry samples in the UKB have between 3,150 and 13,037 (average 10k) total samples for our 75 QC-validated traits. Applying our PRM-test using the European Ancestry PRS to the Non-European target, 71 traits passed our PRM-QC. For these 142 tails we calculated the replication accuracy and replication association as described above.

NEED: ***From Beatrice*** Genotype and phenotype QC.

All of US

All Of Us in biobank contains XX binary and XX continuous traits from individuals of multiple ancestries. Trait matches were found with an initial database search and then were inspected by eye to determine if trait descriptions were sufficiently similar. This process yielded 32 traits, 25 that passed our PRM-QC. These traits have between 8595 and 127677 observations (average 73k) in the European ancestry samples. For these 50 tails the replication accuracy and replication association was calculated as described above.

Genome-wide association analyses and PRS estimation European UKB samples were split randomly into 50% base and 50% target data sets. GWASs were applied to the base data using PLINK-1.9 (6) applying a linear additive model to the standardised trait values. PRScice-2(v2.3.4) (26) was applied to the GWAS summary statistics of each trait with high-resolution scoring to select the most predictive P-value threshold **NEED: using the target data**. Individual risk scores were calculated for UKB samples in the target data.

Data Analysis

The PRS-regression tail test

The PRS-test first applies a regression of PRS on trait values where both the PRS and trait are standardised to have standard deviation of 1. For each trait, the residuals from the regression are then split into quantiles according to the trait distribution and two-tailed t-tests are applied to test whether the residuals are significantly different to zero. The tests were applied to individiuals in the top and bottom 1% of the trait distribution. The effect size in each tail is defined such that a positive effect is indicative of regression to the mean in both the upper and lower tails as follows

$$\frac{1}{n}\sum_{i}^{n}|f(y_i)| - |PRS_i|$$

where $f(y)$ is the predicted PRS from the regression given trait value y and n is the number of observations.

To avoid false positive associations arising from unusual trait distributions, the residuals of samples in the middle of the trait distribution were tested for departure from the null of zero mean. Specifically, for each trait, two-sided t-tests for zero mean were applied to the residuals assigned in each of the middle 80 centiles of the trait distribution. Traits were deemed to fail QC if the smallest P-value was significant at a Bonferroni-corrected level of 0.05 (P $\leq$ 0.05/80). A robust empirical P-value for deviation in trait tails was calculated as follows: for each centile of trait one-sided t-tests for residuals less than and greater than zero were calculated. The P-value for regression to the mean in the upper tail was calculated from the rank of the tests for residuals significantly less than zero and the lower tail empirical P-value from the greater than zero tests. The tests were performed on all 1,121 continuous traits in UKB.

Trumpet plots

Trumpet plots show the effect sizes and MAF of independent genome-wide significant variants. Common variants (MAF>1%) were selected as the most significant variant (P$\leq 5\times10^{-8}$) in 1Mb windows from publicly available GWAS summary statistics generated using UKB samples of European ancestry (14). Exome sequencing associations from the analysis of UKB European ancestry samples were downloaded (15). Results are plotted for the most significant deleterious, missense or predicted loss of function variants in each gene (P$\leq 2.18\times10^{-11}$, one association per gene). Genome-wide significant burden associations are shown where no single variant within the gene is genome-wide significant.

Sibling concordance tests The conditional distribution under polygenicity of an individual's trait value given their siblings trait value is predicted by theory (19)

$$p(s_2 \mid s_1) \sim \mathcal{N}\left(\frac{1}{2}s_1h^2, 1-\frac{h^4}{4}\right) \quad \textbf{(1)}$$

where s_1 and s_2 denote the index and conditional-sibling trait values respectively and h^2 is the trait heritability. Rare large-effect variants can lead to deviations from this distribution, particularly in the tails. Specifically, rare large-effect variants within families, *Mendelian effects*, will result in excess concordance between siblings when they both inherit the effect variant. Conversely, a large effect *de novo* mutation in one sibling will result in excess discordance. The sibling test for *Mendelian* effects applies a binomial test for a greater than expected number of sibling pairs both lying in a defined quantile of the tail of the trait distribution by calculating the following z-statistic

$$z = \frac{r - n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}}$$

where n is the number of index siblings in the quantile q under test, r is the number of siblings pairs both in q and π_0 is the expected proportion of sibling-pairs in the quantile under polygenicity

$$\pi_0 = \frac{1}{n}\sum_{i\in A_q}\left(1-\Phi\left(\frac{\Phi^{-1}(q)-\frac{s_1^{(i)}h^2}{2}}{\sqrt{1-\frac{h^4}{4}}}\right)\right)$$

where A_q is the set of index siblings in trait quantile q, Φ is the normal normal cumulative density function, Φ^{-1} is its inverse and n is the number of sibling pairs.

The sibling test for *de novo* effects constructs a z-statistic for the distance between index siblings in the tail of the trait distribution and their conditional siblings as follows

$$z = \frac{\sum_{i\in A_q} s_2^{(i)} - \frac{1}{2}s_1^{(i)}h^2}{\sqrt{n(1-\frac{h^4}{4})}}$$

For index siblings in the lower and upper tails the alternative hypotheses of $H_1 : z > 0$ and $H_1 : z < 0$ are tested respectively. For further details see Souaiaia et al (19). Heritability h^2 required for the *Mendelian* and *de novo* tests is estimated from the sibling pairs by taking the maximum likelihood of h^2 under the null hypotheses of both tests

$$\log \mathcal{L} \propto -n \log\left(1 - \frac{h^4}{4}\right) - \frac{1}{1 - \frac{h^4}{4}} \sum_{i=1}^{n} \left(s_2^{(i)} - \frac{s_1^{(i)} h^2}{2}\right)^2$$

The sibling meta P-value was calculated by combining P-values from the *Mendelian* and *de novo* tests using Fisher's method to give a χ^2 statistic with 4 degrees of freedom,

$$\chi_4^2 = -2(\log p_{Mendelian} + \log p_{denovo})$$

Selection Analyses

Relative Lifetime Reproductive Success

The number of offspring is recorded in the UKB for both females and males, fields *2734* and *2405* respectively. We followed the methodology of Beauchamp (16) and later applied by Sanjak et al (2) which involves splitting samples into four birth cohorts (1934-42, 1943-48, 1949-55, 1956-65) and dividing the number of offspring of each individual by the cohort mean to calculate relative lifetime reproductive success (Y_{rLRS}). The pressence of selective pressure acting on the traits is assessed by fitting the following regression model to relative lifetime reproductive success (2)

$$Y_{rLRS} = \beta_0 + \beta z + \gamma z^2 \tag{2}$$

where z denotes the residualized and normalized trait values and β and γ are linear and quadratic regression parameters. The pressence and type of selection is inferred from the regression parametrers as folows:

- If the P-value associated with the quadratic parameter γ is significant (FDR 5%), then:
  - If $\gamma < 0$, the trait is classified as under stabilizing selection. (i.e. negative selection both tails).
  - If $\gamma > 0$, the trait is classified as under disruptive selection (i.e. negative selection is not occuring in either tail).
- If only the p-value associated with the linear parameter β is signficant (FDR 5%), then:
  - If $\beta < 0$, trait is classified as under negative directional selection (i.e. negative selection in the upper tail).
  - If $\beta > 0$, trait is classified as under positive directional selection (i.e. negative selection in the lower tail).
- If neither parameter is nominally significant, trait is classified as under neutral selection (i.e negative selection is not occuring in either tail.

Measuring Fitness in the Tails

The following proxy measures of fitness were tested for association with PRM in tails:

- **Adjusted number of illnesses:** The number of self-reported non-cancer illnesses (UKB trait Field 2002) is standardized by applying a regression model with covariates for age, sex, recruitment centre and obtaining residuals.
- **Number of miscarriages:** Trait Field 3829.
- **Number of still-births:** Trait Field 3839.
- **Paternal Age:** Paricipant age (Field 21022) is substracted from father's Age (Field 2946) to give the paternal age at birth.

Association between these measures and PRM was calculated as follows. For each trait, the mean value of the four proxies of selection are calculated for individuals in the upper and lower 1% of the trait distribution separately. Association is then measured by Spearman's rank correlation between the mean tail values for each proxy and the PRM effect size across all trait tails.

Simulating Stabilizing Selection using SLiM

Using SLiM v4.0 (1), we simulated a quantitative polygenic trait under a standard Wright-Fisher (WF) model of stabilising selection. First a population of N=10,000 diploid individuals of 1Mb was simulated to equilibrium (10N generations) with a constant mutation of 2.36×10^{-8} per base pair (bp) and recombination rate of 10^{-8} per bp under neutrallity. Mutations were assigned effect sizes simulated from a gamma distribution with shape parameter 0.186 and mean 0.01026 and subsequently multiplied by -1 with probability 0.5 to give a symmetrical effect size distribution. This distribution has been used previously to simulate genetic effects in humans **NEED: ref** and reflects a prior belief in a small number of large effects and many small effects **NEED: ref**. Individual trait values were determined by the sum of all genetic effects.

After neutral genetic diversity was achieved (10N generations), stabilising selection on the trait was introduced for 1N generations through a Gaussian fitness function

$$f_t(x) = 1 + \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} \tag{3}$$

where x is the phenotype of an individual at generation t, μ and σ are the trait mean and standard deviation at the start of selection (i.e., $t = 10N$). The number of offspring each individual **NEED: contributed to the next generation was...** Stabilizing selection was simulated for a further 1N generations after the 10N burn-in generations.

Simulation data was collected by logging, at each generation, the phenotypic mean and standard deviation of the population, statistics of mutations present in the populations, including allele frequencies and effect sizes, and the mutational profile of each individual. To investigate the contributopn of low-frequency variants and mimic a PRS based on common variants, a second genetic value was calculate using only variants with $MAF > 1\%$.

Replication

Bootstrap Analysis using Target Subsets

Our 75 QC-validated traits were required to have at least 100k total samples. In the target sample (half the total), after outlier removal our QC-validated traits had between of 50,085 and 185,362 (average 142k) target samples. To explore whether our PRM-test results replicate with smaller sample sizes, we repeatedly sampled subsets of between 5k and 50k target samples and performed our PRM-test on these subsets. For each value between 5k and 50k (counting by 5k) we performed 100 trials across each of the 150 trait tails and calculated the **(1)** relative accuracy (true and false positives and negatives) relative to our full dataset (FDR 5%) and **(2)** the linear and the rank correlation of for the regression effect size across all the tails.

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Supplement

Addendum 1: PRS Prediction

In this manuscript we have focused on polygenic regression to the mean, which can be observed when plotting the average PRS score (Y) across the quantiles of the trait distribution (X) (see ***Figure 6***a). Importantly, when we visualize the relationship between the average phenotype (Y) across the quantiles of the PRS distribution the effect isn't visible in mean (see ***Figure 6***o) and can only be observed in variance (see ***Figure 6***r) because non-polygenic influences on the trait appear independent from the PRS distribution*** and thus are uniformly distributed. (*THIS IS IMPORTANT -suggests non-additive).

Here we provide an example of the effect that PRM can have on disease prediction using the continuous trait RBC distribution width (see ***Figure 6***x), namely that, a PRS score in the bottom 5% of the PRS increases the probability of being in the bottom 50%, bottom 25%, bottom 5%, bottom 1%, and bottom 0.1% of the trait distribution by 1.4, 1.8, 2.8, 3.2, and 3.5 times. Consider the other side of the trait distribution, a bottom 5% PRS score reduces the probability of trait score in the top 50% (0.595), top 25% (0.495), top 5% (0.62), and top 1% (0.855) but does not reduce the probability of being in the top 0.1% (1.2x).

Continuing, a PRS score in the top 5% of the PRS, increses the risk of being in the top 50% (1.4), top 25% (1.7), top 5% (1.78), top 1% (1.4x), but does not increase the risk of being in the top 0.1% (0.816).

To summarize, for this trait the predictive value of polygenic scores is:

- Low PRS increases probability of very low trait scores
- Low PRS descreases probability of moderately high trait scores
- High PRS increases probability of moderately high trait score but not very high trait score

Consequently, if disease state occurs for the top 0.1% of the trait distribution, PRS is not predictive.

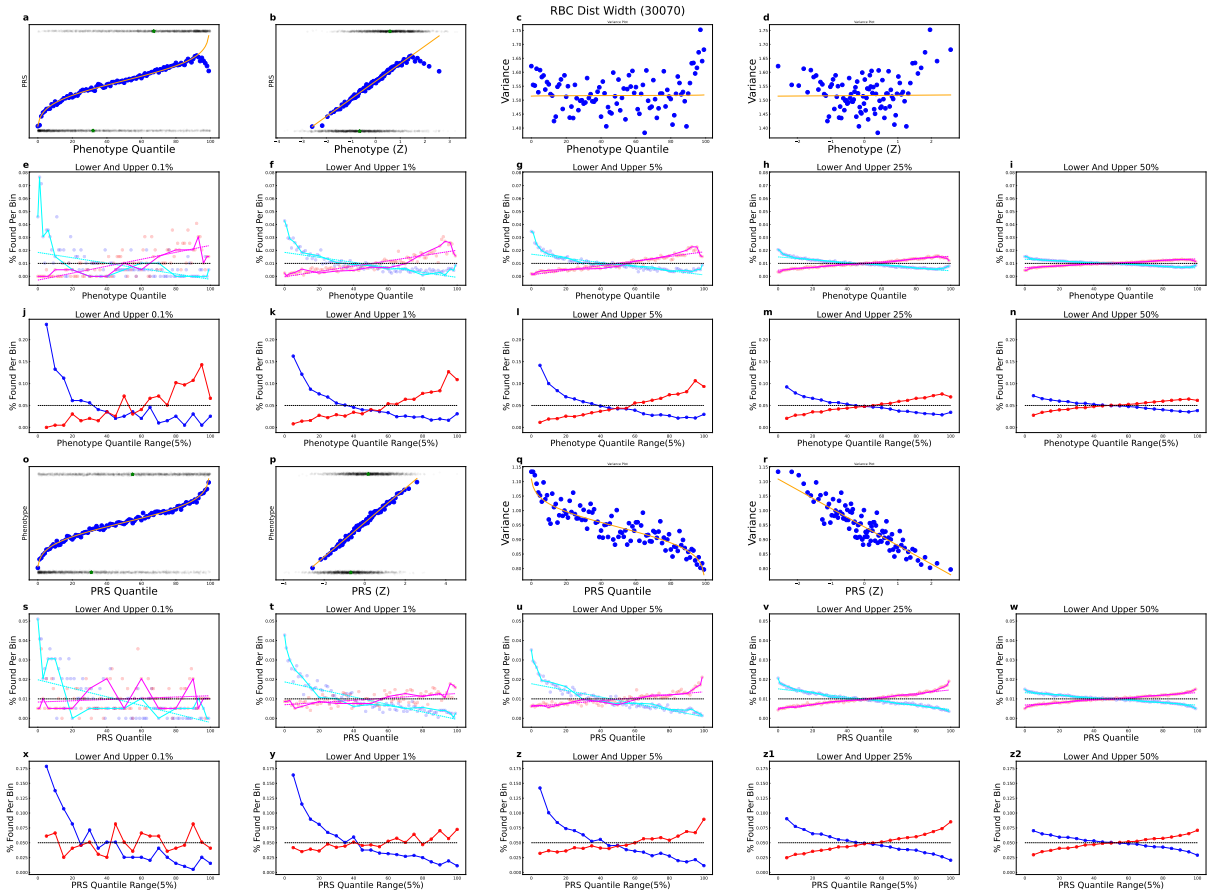

Fig. 6: **Sample and SNP Workflow**

Addendum 2: QC Figures

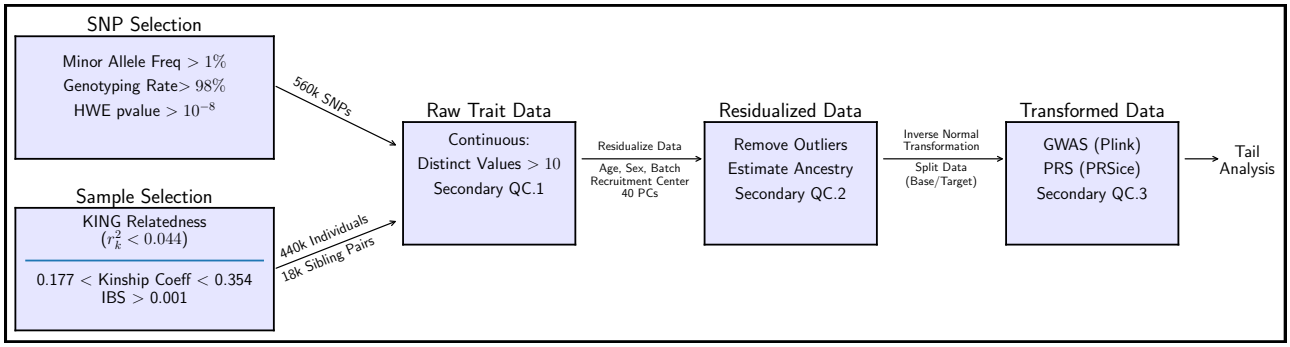

Fig. 7: **Sample and SNP Workflow**

Filter	Classification	Criteria	Where Applied	Result [Running]
Unique	Unique/Redundant	No Genetic Duplicates ($r_g < 99\%$)	Inverse Transfomed Data	(133,131)
Continuity	Continuous/Semi-Cont	Top Two Model Values < 50% of data	Raw Trait Data	(247,17), [122,11]
Normality	Normal/Alternative	Unimodal (Dip Test > 5%), Symetric ($\gamma < 1.5$)	Residualized Data	(202,62), [92,30]
Additivity	Polygenic/Oligogenic	max(Variant r^2) < 5%	GWAS Data	(261,3), [91,1]
Heritability	Heritible/Noisy	SNP $h^2 > 5\%$, PRS(r^2) > 0.5%	Inverse Transformed Data	(167,97), [75,16]

Fig. 8: Secondary QC

Addendum 3: Lifetime Reproductive Success

Trait	All Samples		European Samples		Sanjak et al					
	Linear Parameter (β)	Quadratic Parameter (γ)	Linear Parameter (β)	Quadratic Parameter (γ)	Aggregate Result		Males		Females	
					β	γ	β	γ	β	γ
Bmi	0.05* (positive)	-0.028* (stabilizing)	0.058* (positive)	-0.048* (stabilizing)	positive	stabilizing	0.072*	-0.0327*	0.0601*	-0.0327*
Basal Metabolic Rate	0.045* (positive)	-0.019* (stabilizing)	0.083* (positive)	-0.036* (stabilizing)	positive	stabilizing	0.0747*	-0.03*	0.0405*	-0.03*
Birth Weight	0.011* (positive)	-0.007* (stabilizing)	0.019* (positive)	-0.002	positive	stabilizing	0.0208*	-0.0016	0.0047	-0.0016
Body Fat Percentage	0.019* (positive)	-0.019* (stabilizing)	0.01* (positive)	-0.032* (stabilizing)	positive	stabilizing	0.0198*	-0.0273*	0.0188*	-0.0273*
Fathers Age	-0.074* (negative)	0.005	-0.074* (negative)	0.005	negative	diverging	-0.058*	0.0004	-0.0744*	0.0004
Fluid Intelligence Score	-0.029* (negative)	0.008* (diverging)	-0.02	0.007	positive/negative	diverging	0.0139	0.0009	-0.0473	0.0009
Forced Expiratory Vol	0.016* (positive)	-0.007* (stabilizing)	0.023* (positive)	-0.007* (stabilizing)	positive	stabilizing	0.0335*	-0.0109*	0.0126	-0.0109*
Forced Vital Capacity Fvc	0.018* (positive)	-0.009* (stabilizing)	0.023* (positive)	-0.011* (stabilizing)	positive	stabilizing	0.0323*	-0.0141*	0.0165*	-0.0141*
Hand Grip Strength Left	0.044* (positive)	-0.002	0.044* (positive)	-0.001	positive	stabilizing	0.0728*	-0.0023	0.0124*	-0.0023
Hand Grip Strength Right	0.038* (positive)	-0.001	0.038* (positive)	-0.001	positive	stabilizing	0.0656*	-0.0026	0.0047*	-0.0026
Heel BMD	0.006	-0.013* (stabilizing)	0.006	-0.013* (stabilizing)	positive/negative	stabilizing	0.0325*	-0.0094*	-0.0105	-0.0094*
Heel BMD T Score	0.006	-0.013* (stabilizing)	0.006	-0.013* (stabilizing)	positive/negative	stabilizing	0.0314*	-0.0092*	-0.0104	-0.0092*
Heel Qui	0.006	-0.013* (stabilizing)	0.006	-0.013* (stabilizing)	positive/negative	stabilizing	0.0314*	-0.0092*	-0.0104	-0.0092*
Height	-0.006* (negative)	-0.014* (stabilizing)	-0.006* (negative)	-0.014* (stabilizing)	positive/negative	stabilizing	0.0229*	-0.0152*	-0.0283*	-0.0152*
Highest Edu	-0.068* (negative)	0.013* (diverging)	-0.068* (negative)	0.013* (diverging)	positive/negative	diverging	0.0086	0.0313*	-0.0612	0.0313*
Hip Circumference	0.034* (positive)	-0.022* (stabilizing)	0.034* (positive)	-0.022* (stabilizing)	positive	stabilizing	0.0505*	-0.0278*	0.0357*	-0.0278*
Maximum Digits Remembered	-0.004	-0.01* (stabilizing)	0.016	-0.01* (stabilizing)	negative	stabilizing	-0.0011	-0.0014	-0.0248	-0.0014
Time to Identify Matches	-0.021* (negative)	-0.005* (stabilizing)	-0.042* (negative)	-0.007* (stabilizing)	negative	stabilizing	-0.0434*	-0.0002	-0.0101*	-0.0002
Neuroticism Score	-0.018* (negative)	-0.01* (stabilizing)	-0.038* (negative)	-0.012* (stabilizing)	negative	stabilizing	-0.0318*	-0.0067	-0.0003	-0.0067
Pulse Rate	-0.022* (negative)	-0.011* (stabilizing)	-0.022* (negative)	-0.011* (stabilizing)	negative	stabilizing	-0.0316*	-0.0062*	-0.0136*	-0.0062*
Waist Circumference	0.053* (positive)	-0.029* (stabilizing)	0.053* (positive)	-0.029* (stabilizing)	positive	stabilizing	0.0459*	-0.0331*	0.0649*	-0.0331*
Weight	0.047* (positive)	-0.028* (stabilizing)	0.07* (positive)	-0.047* (stabilizing)	positive	stabilizing	0.0761*	-0.0343*	0.0439*	-0.0343*

Fig. 9: **Comparison of Selection Estimates**